



Aquatic Knee Stabilizer for Spinal Cord Injury Rehabilitation

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Background

- Incomplete spinal cord injury (SCI) patients struggle to maintain standing posture during aquatic therapy
- Caregivers must provide manual knee support
- Limits progress & efficiency
- Existing braces are unsuitable for water use

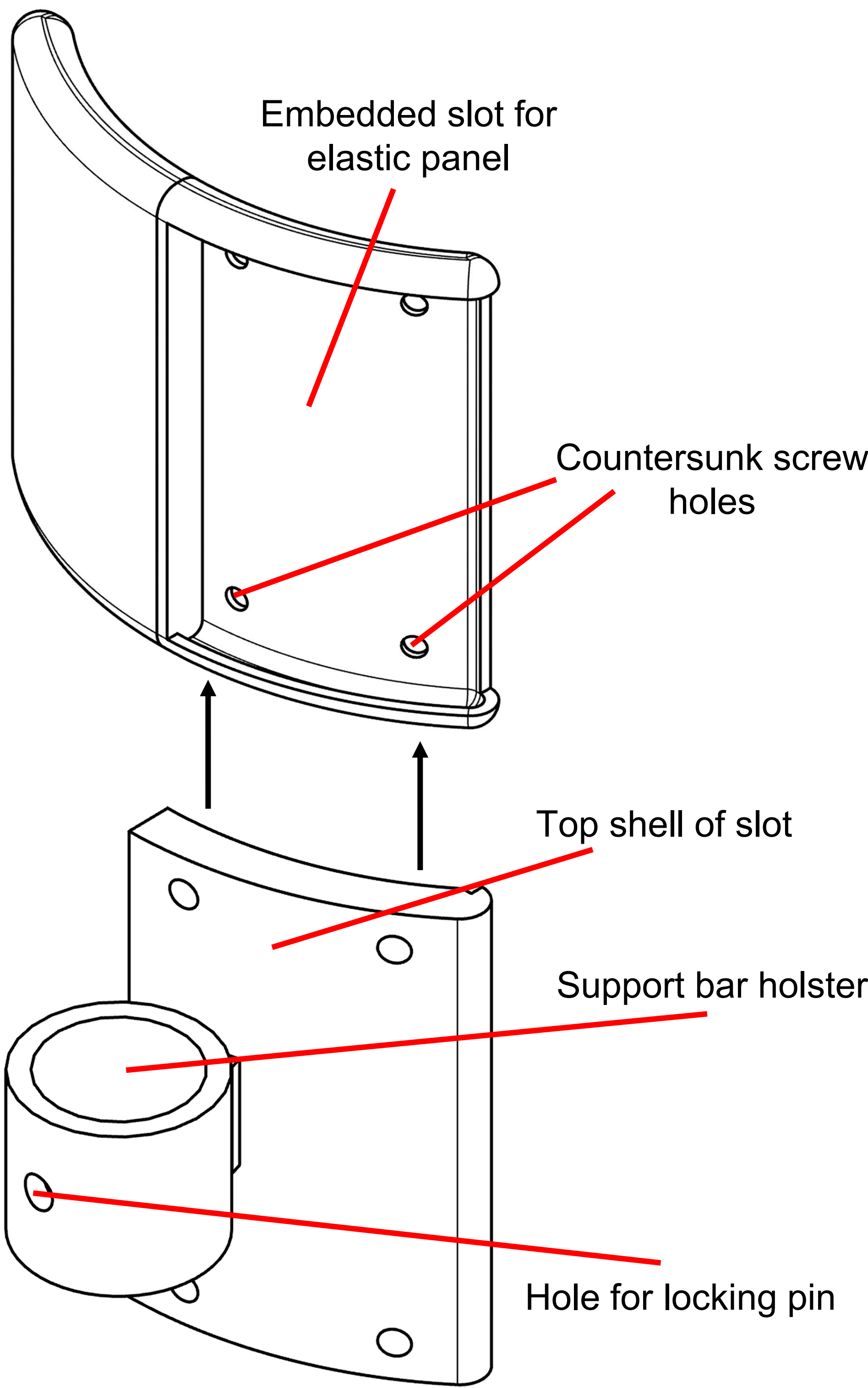


Caregiver manually locking patient's knees

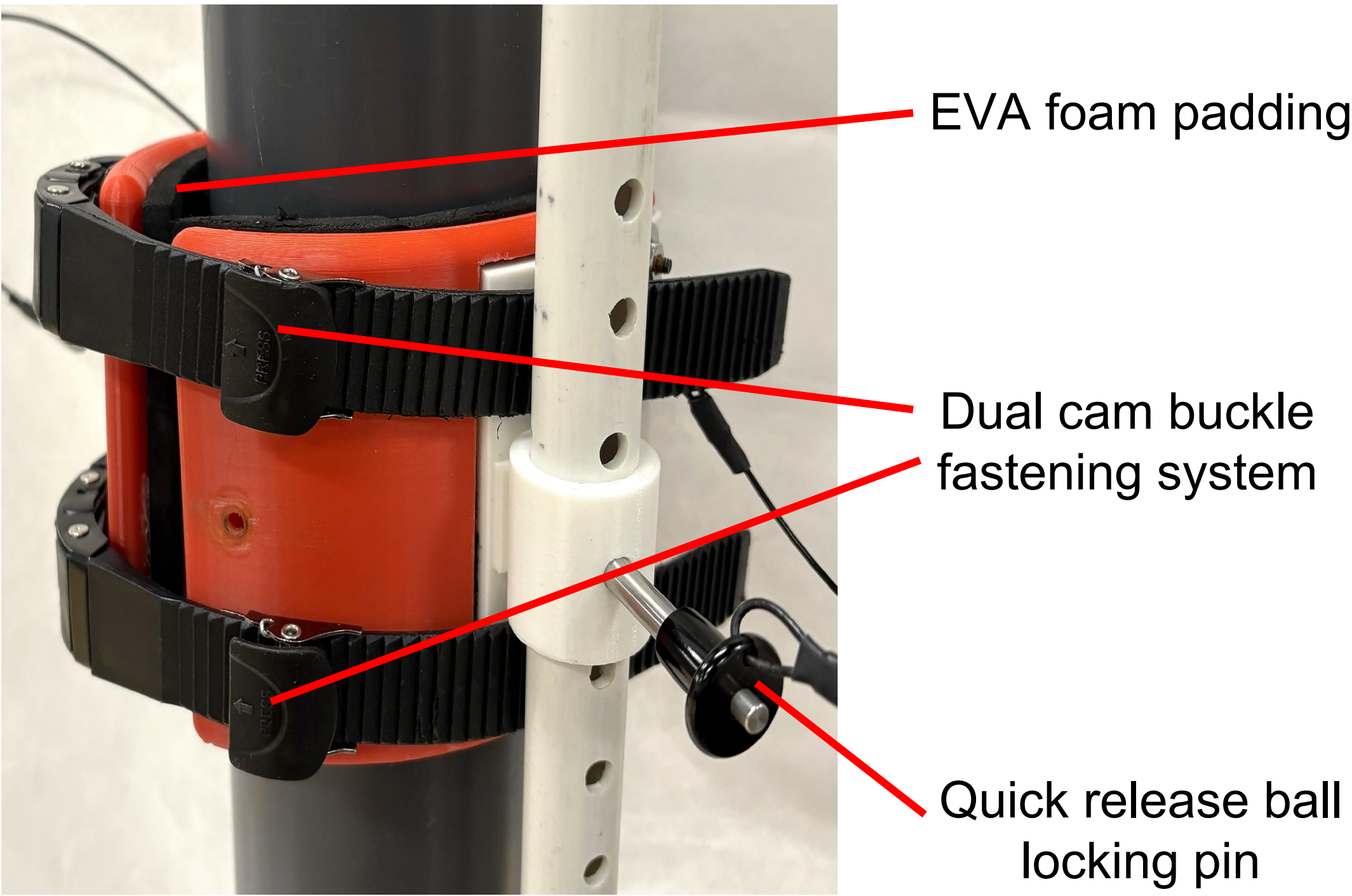
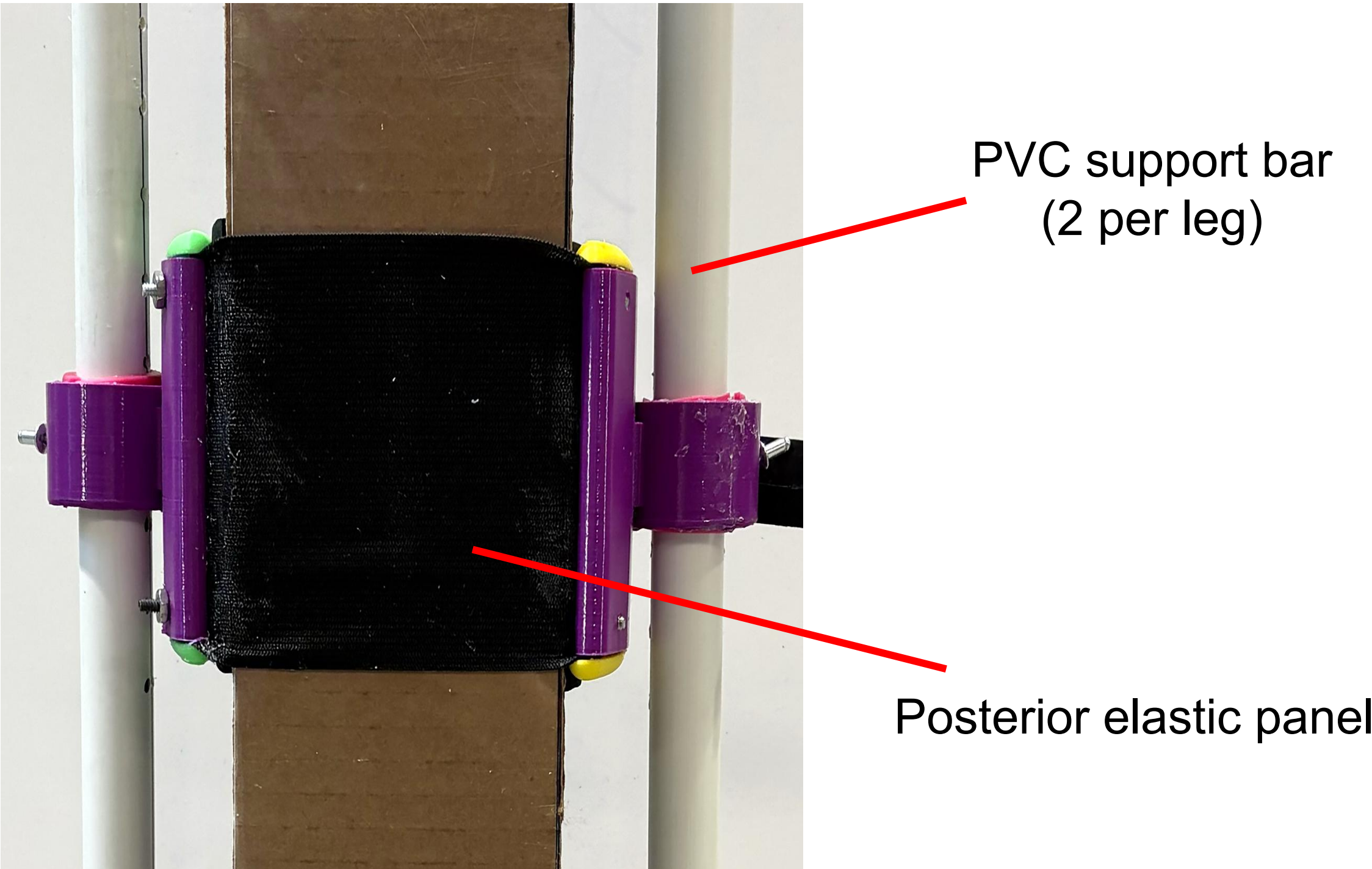
Solution Preview

- Supports and immobilizes the knee from mid-thigh to mid-calf (stabilization)
- Constructed from waterproof, chlorine-resistant materials (durability)
- Implements modular design (adjustability and fit)
- Lightweight cuffs and removable support bars (ease of use)

Final Prototype



CAD Drawing of Cuff Components



Assembled Prototype (Posterior and Isometric Views)

Design Criteria and Testing

#	Design Criterion	Testing Method	Passing Criteria	Results
1	Stabilization	4 non-SCI (land); measure knee flexion angle while squatting	Avg. flexion $\leq 25^\circ \pm 2^\circ$	Avg. flexion 38.2° ($> 25^\circ$ limit) —> FAIL
2	Ease of Use—Cuffs on Land	3 participant pairs donning & doffing cuffs; record time, 5-point scale user rating	≤ 15 s per cuff (avg); ≤ 1 error; rating ≥ 4	Avg. 14.1 s per cuff; 0 errors; rating 4.33 —> PASS
3	Ease of Use—Support Bars Underwater	5 participant pairs attaching PVC bars; record time, 5-point scale user rating	≤ 30 s per leg (avg); ≤ 1 error; rating ≥ 4	Testing not yet conducted —> PENDING
4	Waterproof & Chlorine Resistance	10 immersion-drying cycles; measure dimensional change & function	No visible dimensional change; no material degradation	No visible material degradation or dimensional change —> PASS

Future Plans

- Re-evaluate stabilization performance in aquatic conditions, as the initial land-based testing represents a more demanding scenario than supported standing in water
- Refine padding design to better match leg anatomy and prevent pressure marks during extended use
- Explore alternative attachment mechanisms for support bars (e.g., clasping system) to enable independent use by patients

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